DAY 10 | POST-READ

React Basics: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
SPA	Loads once, updates parts of the page, fetches JSON from APIs.
Component	A function returning JSX. Capitalised name. One parent element.
JSX	{expressions}, className, self-closing tags, camelCase events, style objects.
Props	Inputs from the parent, read-only. Destructure: function Card({ title }).
children	Whatever sits between a component's tags; great for layouts.
State	const [value, setValue] = useState(initial). Change only via the setter.
Copy, do not mutate	setItems([...items, x]); setUser({ ...user, city }).
Events	onClick, onChange, onSubmit (call e.preventDefault()).
Project setup	npm create vite@latest app -- --template react, npm install, npm run dev.
Codex for React	Plan first, one component per prompt, review keys, state, secrets, raw HTML.

Part B: Going beyond today

1. Updating state from the previous value

```
setCount(c => c + 1);   // safe even when several updates happen quickly
```

2. Lifting state up

When two sibling components need the same data, move the state to their closest common parent and pass it down as props, along with functions to change it. The SmartDesk FilterBar and TicketList do exactly this.

3. Prop types with TypeScript (preview)

Many teams use TypeScript with React to catch wrong props before the app runs.

```
type CardProps = { title: string; priority: "low" | "medium" | "high" };
function TicketCard({ title, priority }: CardProps) { ... }
```

4. React DevTools

Install the React Developer Tools browser extension to inspect components, their props and state while the app runs. It is the fastest way to check what Codex built.

5. Accessibility basics

Use real buttons for actions, labels for inputs and enough colour contrast. Ask Codex to review a component for accessibility, then check its suggestions yourself.

Part C: Check yourself

- Why do component names start with a capital letter?
- What is the difference between props and state?
- Why should you never write count = count + 1 for state?
- Why is setItems([...items, x]) better than items.push(x)?
- Name three things you check when reviewing React code from Codex.

Part D: Practice ideas for the long weekend

- Build a ProfileCard component and render three cards with different props.
- Add a search box to the SmartDesk dashboard that filters tickets by title.
- Ask Codex to explain your dashboard's data flow, then correct anything it gets wrong.

Coming up on Day 11: conditional rendering, lists and keys, loading data with loading, error and empty states, never shipping API keys in front-end code, and React talking to FastAPI.

Yoi shuumatsu wo (良い週末を): have a great weekend, and Happy Deepavali